# The Questions

1.

a)

i) What is lightning surge phenomenon in power systems?

[2]

*Lightning results from the movement of the charge particles in the cloud through ionisation of the atmosphere because of very large and pointed electric field which then meet the electrical apparatus such as transmission conductors and or substation equipments . This is known as lightning.*

ii) How is this surge characterised in network planning, design and operation?

[1]

*Very short duration ( micro seconds) but large current in kA range with fast rise time and relatively slow fall time ( typically – 10 kA with 1.2/50 micro seconds.*

iii) How does the lightning surge impact system operation?

[2]

*Lightling brings short circuit to the system , thus interrupting the operation including damaging the insulation of the equipment upon which it falls*.

iv) What mitigation measures are in place to address the impact of a lightning surge in system operation?

[2]

*Lighning arrestor , lightning protection conductor on the top of the tower helping diverting the large current through alternative path , by passing the eqipmnet winding and insulation.*

b)

i) What is sub-synchronous resonance (SSR) in the context of power system operation?

[4]

*Sub synchronous resonance is a phenomenon where different parts of an electromechanical system interacts at its natural frequency. Usually the damping is less , so response is manifested as oscilaltions. Typically multiple sections of steam turbine mass vibrat when excited by torque produced by induced current in the generator rotor body at the resonance frequency. Usually a fixed series compensated transmission system when connected to generator – such problems arise.*

ii) How sub synchronous torsional interaction (SSTI) is different from sub-synchronous control interaction (SSCI)?

[2]

*SSTI involves mechanical parts of turbine in vibration or resonance. SSCI manifests even without the participation of turbine – like power conveter control and network reactance.*

iii) How has the nature of SSR evolved with more converter interfaced generation (CIG) in electric power systems?

Answers

[2]

*It is now more in electrical resonance without involvement of generator turbine shaft. Converter control action provides a spectrum of frequency which matches with some of the natural frequency of the electrical circuits.*

iv) List two approaches to controlling SSR

[2]

*Torsional damper such as NGH circuits across FC. Power system stabiliser.*

c)

i) What is *transient stability* in the context of classical power system operation?

[3]

*Generally it is the ability of a large generator to stay in synchronism with the grid following a fault. It is the acceleration of the turbine mass needing control. It depends on operating load angles and nature of fault clearing mechanism and speed of action.*

ii) How does one improve the transient stability of a power system?

[2]

*Fast excitation control, selective pole switching , auto reclosing CB, turbine fast valving, HP LP by pass , faster CB for fault clearing are the approaches to improve transient stability.*

iii) How has the concern of transient stability changed in view of more renewable generation in the system?

[3]

*Yes, it has . with more and more renewable generation , inertia due to large rotating mass have dropped making the transnient stability problem different – swing dyanmics have been less critical , but large frequency swing of voltage and currents are now more observed.*

[25 marks]

Answers

2.

a)

i) Why are the rotors of synchronous generators round shaped for the power generation application in thermal and nuclear power plants?

[3]

*Synchronous generators driven by steam turbine are high speed machine typically 3000 /3600 rpm . The generator is driven by turbine which is very efficient at high speed. At such high speed , the centrifugal force is so large, it requires the winding to be housed and strongly reinforced. The round rotor shape offers that option.*

ii) Usually the stator of a synchronous generator is laminated whereas the rotor is solid: explain the reason.

[4]

*At normal operation in steady state , rotor does not get any induced voltage/current by the stator current /flux – so no currents in rotor body . It is not required to restrict the induced circulating current in the rotor as that current does not exist – so having a solid construction is handy and advantagegous from manufacturing and cost considerations. Stator on the other hand carries power frequency current – so magnetic materials used in the stator is subjected to iron losses . Making the stator laminiated make sense as it does not produce that losses and so less heating and more efficiency.Solid rotor also offer damping when subjected to disturbances.*

iii) Synchronous machines have synchronous reactance in the range of 1.00 to 2.0 p.u. During a solid fault at the terminal of a synchronous generator, the initial fault current can be as high as 6-8 times the normal load current despite the large synchronous reactance. Explain the reason with reasonable details.

[4]

*This is due to the sudden change of reactance of the stator equivalent coil. This results from the change of the path of the magnetic path immediately following a fault in the system. The stator fluxes are forced to flow through the surface of the rotor as opposed by the rotor body. This is equivalent to small reactance of the stator temporarily which gives rise to large current for few seconds.*

b) A round rotor synchronous generator is connected to the grid at 23 kV (line to line) voltage. The rated power of 500 MW, power factor is 0.85 (lagging) . The synchronous reactance is 1.25 p.u. on its rated base of 588 MVA. The resistance is negligible. Find out:

*This is obtained by solving circuit equation and power angle relationship. I have written few lines of codes in Matlab that produced the following results. In reality it is solving some simple real and reactive power equations involving machine terminal voltage and current that they can easily obtain with their calculator.*

```
pfact = 0.85; pf_angle= -acos(pfact) rated_MW = 500; rated_kV = 23;
rated_amp = rated_MW/(sqrt(3)*rated_kV*pfact); phasor_amp =
complex(rated_amp*cos(pf_angle),rated_amp*sin(pf_angle)); jay =
sqrt(-1); X_s = 1.25; E_fd = rated_kV+sqrt(3)*jay*phasor_amp*X_s;
```

*E_fd_abs= abs(E_fd); delta= rad2deg(angle(E_fd)) S = sqrt(3)*rated_kV*conj(phasor_amp); P_t = real(S); Q_t = imag(S); stab_margin = 90-delta; E_fd2 = 1.1*E_fd_abs; delta_new = rad2deg(asin(rated_MW/(E_fd2*rated_kV))); stab_margin_new = delta-delta_new; Answer_var = [rated_amp; E_fd_abs; delta; P_t; Q_t; abs(S);stab_margin;delta_new;stab_margin_new]*

i) the rated current in kA

[2]

*14.76 kA*

ii) the excitation (line to line) voltage in kV

[3]

*48.22 kV*

iii) load angle (δ) in degree

[2]

*34.29 degree*

iv) stability margin in degree

[1]

*55.70 degree*

iv) reactive power in MVAR

[3]

*310.00 MVAr*

c) The excitation voltage is increased by 10%

i) Find out the change in stability margin in degree with respect to the rated excitation voltage.

[3]

10 degrees

[25 marks]

Answers

3.

a)

i) What is Park transformation in the context of synchronous machine dynamic modelling ?

[2]

*Park transformation is representation of electrical quantities ( current, voltage, flux) in a rational reference frame. It transforms time varying signals into time invariant signals.*

ii) How does Park transformation overcome the limitation of 'abc' stationary reference frame for circuit analysis?

[4]

*The current/MMF in abc reference frame faces a time varying reluctance because of the movement of the rotor. This makes the inductance time variant to analyse the flux current and voltage current relationship. The park transformation removes this problem by defining the quantitities in abc reference frame in a rotating reference frame. In the rotating reference frame , currents , voltages and fluxes become time invariant and inductance remains constant. It is very convenient to handle these in park reference frame.*

iii) What are stator transients in the context of circuit model of synchronous machine dynamics ?

[3]

*When an AC sinusoidal voltage is impressed across an R-L or R-C circuit , it will have a transient components . With time steady state current results. In synchronous machine stator rotor flux produces speed voltage. There is an additional voltage that arises because of the transformer action when there is dc offset current in the stator in abc reference frame which becomes 50 Hz AC in parke reference frame. This voltage is of 50 Hz and in rotating reference axes.*

iv) Why are such transients neglected in transient stability studies?

[3]

*Including them make the analysis complex. They are of short duration compared to electromechncial swing. Their magnitude is less . So neglecting them simplifies the analysis.*

v) What will be the impact of the inclusion of stator transients on the generator's dynamic behaviour ?

[3]

*Usually magnitude of this voltage is small compared to speed voltage. In electromechanical stability study problem , they are also fast. Including them will have a 50 Hz component ripple in speed deviation which disappears faster. They offer additional damping in the first 1-2 cycles of power frequency. Their exclusion always give conservative results which is good from the qualitative analysis of stability problem.*

Answers

b)

i) What are the basic functional and constructional differences between the Type 2 (WRIG) and Type 3 (DFIG) wind turbine generators ?

[4]

*Both uses wound rotor . Type 2 uses a bank of resitor through slip ring. It dissipate the slip power. Type 3 converts the slip frequency current/voltage into DC voltage which then feeds an inverter. The slip power is recovered. Type2 can only operate at more than synchronous speed where the current is always coming out of the rotor. In Type 3 currents are in both direction , so allows the operation at below synchronous speed. Type 3 in that sense offers a wide variation of speed range. Type 3 is more efficient over Type 2.*

ii) What is Tip Speed Ratio (TSR) in the context of wind turbine operation ?

[3]

*Tip-speed ratio (TSR) is defined as the ratio between the incoming wind speed and the blade tip speed. Mathematically, it is expressed as:*

$$\lambda = \frac{\omega_m R}{v_w}$$

*Where $\omega_m$ is the generator mechanical speed, R and $v_w$ denote blade length and upcoming wind speed, respectively.*

iii) How does TSR influence the power coefficient (Cp) of a wind turbine ?

[3]

*TSR is very important factor as together with angle of attack α, they influence the wind turbine conversion efficiency $C_p$. A turbine is usually designed with an optimal TSR point. If the upcoming wind speed changes, the generator mechanical rotation must be changed according to:*

$$\omega_m = \frac{\lambda_{opt} v_w}{R}$$

[25 marks]

Answers

4.

a)

i) What is the Betz limit in wind power conversion?

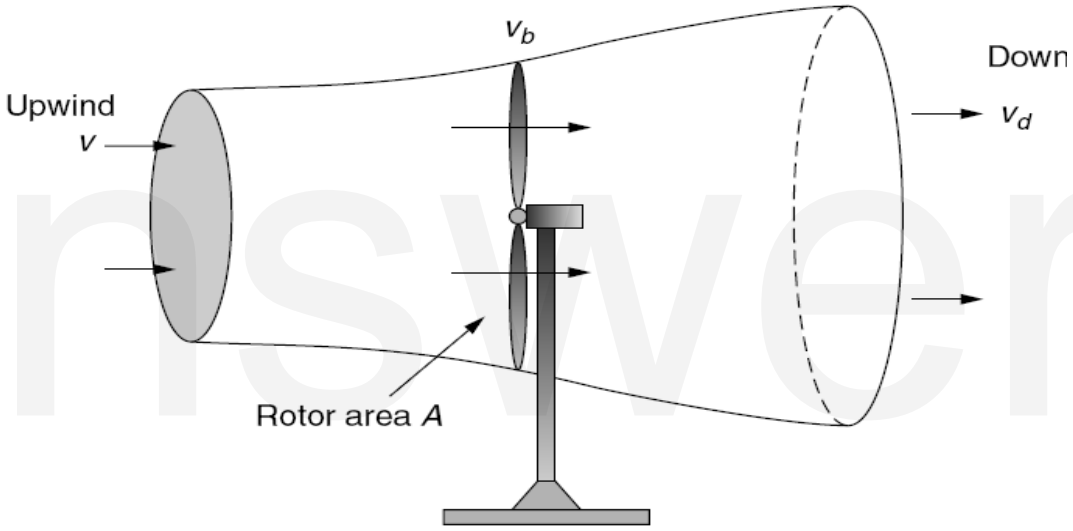
[2]

*It is the upper limit of power that can be obtained in mechanical form from the maximum available power in the wind passing through blade. It is 59.3%calculIt was first computed by Betz , so named after him.*

ii) Starting from the first principle of aerodynamics, establish that the Betz limit is 16/27.

[7]

*Look at the figure below*




*Suppose the rotor swept area is A, the upwind speed is $v$, the downwind speed is $v_d$, the average wind speed at blades is $v_b = \frac{v+v_d}{2}$, and the mass flow of air is $\dot{m}$*

*The power extracted by the blade is*

$$P_b = \frac{1}{2}\dot{m}(v^2 - v_d^2)$$

*the mass flow of air can be expressed as*

$$\dot{m} = \rho A v_b$$

*Substituting $\dot{m}$ into $P_b$ yields*

$$P_b = \frac{1}{2}\rho A\left(\frac{v + v_d}{2}\right)(v^2 - v_d^2)$$

*Assume $\lambda = \frac{v_d}{v}$, the expression of $P_b$ becomes*

$$P_b = \frac{1}{2}\rho A\left(\frac{v + \lambda v}{2}\right)(v^2 - \lambda^2 v^2)$$

Answers

$$P_b = \frac{1}{2}\rho A v^3 \left[\frac{1}{2}(1+\lambda)(1-\lambda^2)\right]$$

$$P_b = \frac{1}{2}\rho A v^3 C_p$$

*with $C_p$is equal to turbine efficiency.*

*In order to find the maximum value, take the derivative of $C_p$ with respect to λ and set this to zero*

$$\frac{dC_p}{d\lambda} = \frac{1}{2}[(1+\lambda)(-2\lambda) + (1-\lambda^2)] = 0$$

*After simple calculation, the solution is $\lambda = \frac{1}{3}$. Substituting back the value of λ into $C_p$ results*

$$C_p = \frac{1}{2}\left(1+\frac{1}{3}\right)\left(1-\frac{1}{3^2}\right) = \frac{16}{27}$$

b)

i) What is passive stall in a wind turbine?

[2]

*It is the action of the blade aerofoil structure on the wind hitting the blade. It results in drop in lift forces.*

ii) How is such stall achieved in practice?

[2]

*It is realised through shaping the geometry of the blade such that when increased wind velocity hits the blade , the wind flow on the upper surface of the blade becomes more turbulent thus resulting in less lift forces.*

iii) How does active stall control differ from passive stall control?

[2]

*In active stall control the angle of surface of the blades moves with respect to the resultant direction of the wind.*

iv) Why has the industry discontinued active stall control?

[2]

*It is expensive to implement and it requires additional actuator system. It also impacts the working life of wind turbine.*

v) How is the gearbox modelled for dynamic studies of wind turbines ?

[4]

*The gearbox is modelled as simple gain when everything is represented in actual unit. It is expressed as the ratio of the number of teeth. For dynamic study it is represented as small mass also. For power system stability studies it is represented as a factor which is either unitless or with unit depending on whether per unit system is adopted or not.*

vi) Why are turbine blades not controlled below the rated wind speed ?

[2]

*Controlling them at below wind speed results in sub optimal tip speed ratio and Cp , so less efficient energy conversion.*

Answers

vii) Why does a large off-shore wind farm, located deep in the sea, requires HVDC technology for power transmission ?

[2]

*Transmission at less than 275kV or 400kV is expensive as under sea cable size increase. At high voltage AC , the charging current increases thus leading to large over voltage at the receiving end of the offshore power transmission. Typically one single circuit 400 kV Ac system produces 20 MVAr per km . For a wind farm which is about 100 km into sea from the shore will produce 2000 MVar . It is very difficult to compensate that level of reactive power through static compensation. So HDVC is a solution which can be cost effective.*

[25 marks]

Answers